1. (12%) Determine whether these statements are **true or false**. (圈選True(T)或False(F))
   (a) T F $\subseteq \{$
   (b) T F $\in \{$
   (c) T F $\} \in \{\{ \}\}$
   (d) T F $\} \subset \{$
   (e) T F $\}\} \subset \{ \quad ,\}$
   (f) T F $\{\{ \}\} \subset \{\{\emptyset\},\{\emptyset\}\}$

2. (8%) Determine **whether** each of these sets is the power set of a set **or not**, where a and b are distinct elements. (圈選True(T)或False(F))
   (a) T F $\emptyset$
   (b) T F $\{\emptyset, \{a\}\}$
   (c) T F $\{\emptyset, \{a\}, \{\emptyset, a\}\}$
   (d) T F $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

3. (5%) Let S = {1, 2, 3, 4, 5, {6, 7}}. What is the **cardinality** of the **power set of S**?
   (直接將答案填入) ans:__________

4. Let f, g: $\mathbf{R} \to \mathbf{R}$. If $f(x)=3x$ and $f \circ g(x)=3|x+4|-5$.
   (a) (6%) Find $f^{-1}(x)$. ans:________________
   (b) (6%) Find $g(x)$. ans:________________

5. (8%) What is the **cardinality** of each of these sets? (直接將答案填入)
   (a) $\emptyset$ ans:_______
   (b) $\{\emptyset\}$ ans:_______
   (c) $\{\emptyset, \{\emptyset\}\}$ ans:_______
   (d) $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$ ans:_______

6. (12%) Determine whether each of the following statements is **true or false**. (圈選True(T)或False(F)即可)
   (a) T F $\mathbf{Z} \to \mathbf{Z}$, $f(n) = n-5$ is one-to-one
   (b) T F $\mathbf{Z} \to \mathbf{Z}$, $f(n) = n-5$ is onto
   (c) T F $: \mathbf{N} \to \mathbf{N}$, $f(n) = n+5$ is one-to-one
   (d) T F $: \mathbf{N} \to \mathbf{N}$, $f(n) = n+5$ is onto
   (e) T F $f: \mathbf{Z} \times \mathbf{Z} \to \mathrm{Z}$, $f((m, n)) = \mathrm{m} - \mathrm{n}$. is one-to-one
   (f) T F $f: \mathbf{Z} \times \mathbf{Z} \to \mathrm{Z}$, $f((m, n)) = \mathrm{m} - \mathrm{n}$. is onto

7. (12%) Let A= $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and B= $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$. Find the (a) **join**, (b) **meet**, and (c) **Boolean product** of these 2 **zero-one matrices**.

8. (12%) Find
   (a) $\bigcup_{i=1}^{+\infty} [-\frac{1}{i}, \frac{1}{i}]$
   ans:________
   (b) $\bigcap_{i=1}^{+\infty} (1 - \frac{1}{i}, 1)$
   ans:________
   (c) $\bigcap_{i=1}^{+\infty} [1 - \frac{1}{i}, 1]$
   ans:________

9. (8%)Show that if A, B, and C are sets, then $\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$ **using a membership table.**

10. Determine and explain whether each of these sets is **countable(可數) or uncountable(不可數)**. If your answer is **countable**, exhibit a **one-to-one correspondence** between the set of positive integers($\mathbf{Z}^{+}$) and that set. If the answer is uncountable, prove it!
   (a) (8%) The set of Integers that are greater than 10.
   (b) (8%) The set of positive real numbers that are less than 2 and with decimal representations contain only 1s or 9s. (e.g., 1.11, 1.191, …).